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Inventor(s)

Krugel; Samantha J. et al.

AUDIO OUTPUT DEVICE WITH A LIGHT EMITTING UNIT

Abstract

An audio output device may include a housing, a speaker provided in the housing, a plurality of actuators provided on the housing, and a light emitting unit provided on the housing. The light emitting unit includes a plurality of light emitting segments arranged in a line. The audio output device may also include a controller located within the housing and including an electronic processor, a memory, and a transceiver. The controller is coupled to the speaker, the plurality of actuators, and the light emitting unit. The controller is configured to determine that a first actuator of the plurality of actuators is actuated for a first predetermined amount of time, and operate the light emitting unit according to a first light sequence in response to the first actuator being actuated.

Inventors: Krugel; Samantha J. (Wauwatosa, WI), McCormick; Chelsea L. (Oak Creek, WI), Ogden; Matthew H. (Milwaukee, WI), Hadfield; Jacob D. (Mukwonago, WI), Millane; Christopher R. (Menomonee Falls, WI), Liu; Long Biao (Dongguan City, CN)

Applicant: MILWAUKEE ELECTRIC TOOL CORPORATION (Brookfield, WI)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/557,689, filed Feb. 26, 2024, the entire contents of which are incorporated by reference herein.

FIELD OF THE INVENTION

[0002] The present invention relates to an audio output device and, more particularly, a light emitting unit for an audio output device.

SUMMARY OF THE INVENTION

[0003] In addition to playing content, such as music, podcasts, audio files, etc. provided from an external device, audio output devices may convey information to a user. Audio output devices may include a light emitting unit that provides lighting schemes specific to certain information to be conveyed. For example, the light emitting unit may illuminate in a certain scheme to convey that the audio output device is successfully paired via Bluetooth® to an external device, such as a mobile phone. It may be advantageous for the light emitting unit of the audio output device to illuminate in various schemes that correspond to various inputs on the audio output device.

[0004] In some aspects, the techniques described herein relate to an audio output device including a housing; a speaker provided in the housing; a plurality of actuators provided on the housing; a light emitting unit provided on the housing, the light emitting unit including a plurality of light emitting segments arranged in a line; and a controller located within the housing and including an electronic processor, a memory, and a transceiver, the controller coupled to the speaker, the plurality of actuators, and the light emitting unit, the controller configured to determine that a first actuator of the plurality of actuators is actuated, and operate the light emitting unit according to a first light sequence in response to the first actuator being actuated.

[0005] In some aspects, the techniques described herein relate to an audio output device including a housing; a speaker provided in the housing; a plurality of actuators provided on the housing; a light emitting unit provided on the housing, the light emitting unit including a plurality of light emitting segments arranged in a line; and a controller located within the housing and including an electronic processor, a memory, and a transceiver, the controller coupled to the speaker, the plurality of actuators, and the light emitting unit, the controller configured to determine that a first actuator of the plurality of actuators is actuated to initiate an operation, operate the light emitting unit according to a first light sequence in response to the first actuator being actuated and the operation being initiated, and operate the light emitting unit according to a second light sequence in response to the operation being completed, the second light sequence being different than the first light sequence.

[0006] In some aspects, the techniques described herein relate to an audio output device including an elongated housing having a first end, a second end opposite the first end, and a longitudinal axis extending through the first end and the second end; a speaker provided in the elongated housing; a plurality of actuators provided on the elongated housing; and a light emitting unit provided on the elongated housing and including a plurality of light emitting segments arranged in a line extending parallel to the longitudinal axis, the light emitting unit operable to emit a light sequence in response to actuation of one of the plurality of actuators.

[0007] Other aspects of the invention will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 illustrates an audio output device, according to some embodiments.

[0009] FIG. 2 illustrates a user interface of the audio output device of FIG. 1, according to some embodiments.

[0010] FIG. 3 illustrates the user interface and a light emitting unit of FIG. 2, according to some embodiments.

[0011] FIG. 4 is a communication network for the audio output device of FIG. 1, an external device, and an additional audio output device, according to some embodiments.

[0012] FIG. 5 is a block control diagram of the audio output device of FIG. 1, according to some embodiments.

[0013] FIG. 6 is a block diagram of an external device, according to some embodiments.

[0014] FIG. 7A is a flow chart illustrating a method of powering ON the audio output device, according to some embodiments.

[0015] FIG. 7B is a flow chart illustrating a method of pairing the audio output device with an external device upon powering ON, according to some embodiments.

[0016] FIG. 8 is a flow chart illustrating a method of powering OFF the audio output device, according to some embodiments.

[0017] FIG. 9A is a flow chart illustrating a method of pairing the audio output device with an external device, according to some embodiments.

[0018] FIG. 9B is a flow chart illustrating a method of the audio output device outputting content from the external device, according to some embodiments.

[0019] FIG. 10 is a flow chart illustrating a method of the audio output device losing a Bluetooth® connection with an external device, according to some embodiments.

[0020] FIG. 11 is a flow chart illustrating a method of daisy chaining the audio output device with a second audio output device, according to some embodiments.

[0021] FIG. 12 is a flow chart illustrating a method of daisy chaining the second audio output device with a third audio output device, according to some embodiments.

[0022] FIG. 13 is a flow chart illustrating a method of removing the second audio output device from the daisy chain, according to some embodiments.

[0023] FIG. 14 is a flow chart illustrating a method of playback control on the audio output device, according to some embodiments.

[0024] FIG. 15 is a flow chart illustrating a method of volume control reaching a maximum/minimum level on the audio output device, according to some embodiments.

[0025] FIG. 16 is a flow chart illustrating a method of controlling a volume and providing a tone in response to an actuation on the audio output device, according to some embodiments.

[0026] FIG. 17 is a graph illustrating volume control when a tone is provided on the audio output device, according to some embodiments.

DETAILED DESCRIPTION

[0027] Before any embodiments are explained in detail, it is to be understood that the embodiments are not limited in its application to the details of the configuration and arrangement of components set forth in the following description or illustrated in the accompanying drawings. The embodiments are capable of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and

variations thereof are meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.

[0028] In addition, it should be understood that embodiments may include hardware, software, and electronic components or modules that, for purposes of discussion, may be illustrated and described as if the majority of the components were implemented solely in hardware. However, one of ordinary skill in the art, and based on a reading of this detailed description, would recognize that, in at least one embodiment, the electronic-based aspects may be implemented in software (e.g., stored on non-transitory computer-readable medium) executable by one or more processing units, such as a microprocessor and/or application specific integrated circuits (“ASICs”). As such, it should be noted that a plurality of hardware and software based devices, as well as a plurality of different structural components, may be utilized to implement the embodiments. For example, “servers,” “computing devices,” “controllers,” “processors,” etc., described in the specification can include one or more processing units, one or more computer-readable medium modules, one or more input/output interfaces, and various connections (e.g., a system bus) connecting the components.

[0029] Relative terminology, such as, for example, “about,” “approximately,” “substantially,” etc., used in connection with a quantity or condition would be understood by those of ordinary skill to be inclusive of the stated value and has the meaning dictated by the context (e.g., the term includes at least the degree of error associated with the measurement accuracy, tolerances [e.g., manufacturing, assembly, use, etc.] associated with the particular value, etc.). Such terminology should also be considered as disclosing the range defined by the absolute values of the two endpoints. For example, the expression “from about 2 to about 4” also discloses the range “from 2 to 4”. The relative terminology may refer to plus or minus a percentage (e.g., 1%, 5%, 10%, or more) of an indicated value.

[0030] It should be understood that although certain drawings illustrate hardware and software located within particular devices, these depictions are for illustrative purposes only. Functionality described herein as being performed by one component may be performed by multiple components in a distributed manner. Likewise, functionality performed by multiple components may be consolidated and performed by a single component. In some embodiments, the illustrated components may be combined or divided into separate software, firmware and/or hardware. For example, instead of being located within and performed by a single electronic processor, logic and processing may be distributed among multiple electronic processors. Regardless of how they are combined or divided, hardware and software components may be located on the same computing device or may be distributed among different computing devices connected by one or more networks or other suitable communication links. Similarly, a component described as performing particular functionality may also perform additional functionality not described herein. For example, a device or structure that is “configured” in a certain way is configured in at least that way but may also be configured in ways that are not explicitly listed.

[0031] FIG. 1 illustrates an audio output device **100**, according to some embodiments. The audio output device **100** may be a portable, wireless device. In some embodiments, the audio output device **100** may receive a wireless signal from an external device, such as external device **305** (FIG. 4), using Bluetooth®, Bluetooth® low energy (BLE), near field communication (NFC), Wi-Fi, cellular data, or other wireless communication schemes. For example, the wireless signal may include sound data, control inputs, notifications, etc. In some embodiments, the audio output device **100** may transmit audio signals and relay wireless signals from the external device to a second audio output device, such as second audio output device **400** (FIG. 4).

[0032] The audio output device **100** includes a housing **105**. In the illustrated embodiment, the housing **105** is an elongated housing with a battery pack interface **110** provided at a first end **115** of the housing **105** for receiving a battery pack **120**. The first end **115** may also be referred to as a top

or a top portion. Opposite the first end **115** of the housing **10** is a second end **125** of the housing **105**. The second end **125** may also be referred to as a base or a base portion. In some embodiments, the housing **105** is generally cylindrical with the second end **125** including a flat surface for supporting or balancing the audio output device **100**. Additionally, the second end **125** may include features (e.g., hooks, magnets, etc.) for attaching the audio output device **100** to another structure. The housing **10** also defines a longitudinal axis A extending through the first end **115** and the second end **125**. When the audio output device **100** is upright and supported by the second end **125** (as shown in FIG. 1), the longitudinal axis A is a vertical axis. The audio output device **100** may also be laid on its side. In such an arrangement, the longitudinal axis A is a horizontal.

[0033] The audio output device **100** further includes at least one speaker **130**, a speaker grill **135** including apertures **140**, a user interface **145**, and a light emitting unit **150**. The speaker **130**, speaker grill **135**, user interface **145**, and light emitting unit may be provided on a rubber portion **155** provided between the first end **115** and the second end **125** of the housing **105**. The speaker grill **135** may be made of metal or plastic and is configured to inhibit debris from contacting the speaker **130**. The speaker grill **135** includes a plurality of apertures **140** that allow sound to be emitted from the speakers **130**. In some embodiments, the speaker grill **135** and the speaker **130** occupy a surface area of the audio output device in a 1:1 ratio. Alternatively, in some embodiments, the speaker grill **135** and the speaker **130** occupy a surface area of the audio output device in a 3:2 ratio. In some embodiments, the audio output device **100** may include one, two, three, or four speakers **130**. The speakers **130** may include a mid-range speaker, a tweeter, a woofer, a subwoofer, a supertweeter, multiple tweeters, or any combination thereof. In other embodiments, the audio output device **100** may include other types of speakers.

[0034] The user interface **145** includes a plurality of actuators that will be described with respect to FIG. 2. In some embodiments, the light emitting unit **150** is provided between the speaker grill **135** and the user interface **145** along the rubber portion **155** of the housing **105**. In the illustrated embodiment, the light emitting unit **150** is provided next to or directly adjacent the user interface **145**. For example, the light emitting unit **150** may be provided on the right-hand side of the user interface **145**. Alternatively, in some embodiments, the light emitting unit **150** may be provided on the left-hand side of the user interface **145**. In still other embodiments, the user interface **145** and the light emitting unit **150** may be provided on other parts of the housing **105**.

[0035] The battery pack **120** may be interchangeable and used with a plurality of battery powered devices. For example, the plurality of battery powered devices may include power tools such as drills, other audio output devices, lighting devices, or other battery powered devices. In one example, the battery pack **120** may be used to power the audio output device **100** and a power tool (not shown)

[0036] In some embodiments, the battery pack **120** includes one or more lithium-ion battery cells. In other embodiments, the battery pack **120** may be of a different chemistry, for example, nickel-cadmium, nickel-metal hydride, and the like. In some embodiments, the battery cells may include one or more cylindrical cells, one or more pouch cells, one or more prismatic cells, or a combination thereof. In some embodiments, the battery pack **120** may include anywhere in the range of three to six battery cells. In the illustrated embodiment, the battery pack **12** is a 12V battery pack outputting a constant 12V output. In other embodiments, the output voltage level of the battery pack **120** may be different. For example, the battery pack **120** may be a 4V battery pack, and 18V battery pack, 28V battery pack, 40V battery pack, or another voltage. The battery pack **120** may also have various capacities (e.g., 3, 4, 5, 6, 8, or 12 A/hr). The output of the battery cells may be in the range of 3.0-6.0 Ampere-hours (Ah).

[0037] FIG. 2 illustrates the user interface **145** of the audio output device **100**, according to some embodiments. The user interface **145** includes a power actuator **160**, a fuel gauge **165**, a play actuator **170**, a volume increase actuator **175**, a volume decrease actuator **180**, a Bluetooth® actuator **185**, a daisy chain actuator **190**, and a charging port **195**. In some embodiments the

actuators **160, 170, 175, 180, 185, 190** are provided in descending order from the first end **115** to the second end **125** of the housing **105** along the longitudinal axis A, as listed in the preceding sentence. For example, the illustrated actuators **170, 175, 180, 185, 190** are arranged in a line that is parallel to the longitudinal axis A. In some embodiments, the power actuator **160** occupies a greater surface area of the rubber portion **155** than each of the play actuator **170**, the volume increase actuator **175**, the volume decrease actuator **180**, the Bluetooth® actuator **185**, and the daisy chain actuator **190**. The actuators **160, 170, 175, 180, 185, 190** may be capacitive touch buttons, physical buttons, dials, switches, slides, and the like. In some embodiments, the actuators **160, 170, 175, 180, 185, 190** may include an LED that illuminates upon actuation. For example, the LED may shine through a window in the rubber portion **155** that is the shape of the function of the particular actuator. In some embodiments, the charging port **195** is a USB-C charging port that provides power to the battery pack **120**. The charging port **195** may also or alternatively provide power to an external device from the battery pack **120**. In other embodiments, the charging port **195** may be a different type of port.

[0038] The fuel gauge **165** is a separate indicator from the light emitting unit **150** and the actuators **160, 170, 175, 180, 185, 190**. In some embodiments, the fuel gauge **165** is provided adjacent the power actuator **160**. For example, the fuel gauge **165** may be provided on an edge of the power actuator **160** closer to the first end **115** of the housing **105** (e.g., on a top edge of the power actuator **160**). In other embodiments, the fuel gauge **165** may be located elsewhere on the housing **105**. The fuel gauge **165** is configured to display an amount of a charge of the battery pack **120** coupled to the battery pack interface **110**. In the illustrated embodiment, the fuel gauge **165** includes a series (e.g., four) LEDs. In other embodiments, the fuel gauge **165** may have other configurations.

[0039] In some embodiments, actuation of the play actuator **170** controls a play/pause function of content (i.e., audio) playing over the speaker **130** of the audio output device **100** from the external device **305**. Actuation of the play actuator **170** may also skip forward or skip backward through the content. In some embodiments, actuation of the volume increase actuator **175** increases the volume of the content playing over the speaker **130**. In some embodiments, actuation of the volume decrease actuator **180** decreases the volume of the content playing over the speaker **130**. In some embodiments, actuation of the Bluetooth® actuator **185** enters the audio output device **100** into a Bluetooth® pairing mode. In some embodiments, actuation of the daisy chain actuator **190** enters the audio output device **100** into a daisy chaining mode.

[0040] FIG. 3 illustrates the user interface **145** and the light emitting unit **150**, according to some embodiments. In some embodiments, the light emitting unit **150** includes a plurality of light emitting segments **200, 205, 210, 215, 220**. In some embodiments, each light emitting segment **200, 205, 210, 215, 220** corresponds to an actuator of the user interface **145**. For example, the first light emitting segment **200** corresponds to the play actuator **170**, the second light emitting segment **205** corresponds to the volume increase actuator **175**, the third light emitting segment **210** corresponds to the volume decrease actuator **180**, the fourth light emitting segment **215** corresponds to the Bluetooth® actuator **185**, and the fifth light emitting segment **220** corresponds to the daisy chain actuator **190**. As such, in the illustrated embodiment, the light emitting unit **150** includes five light emitting segments. In other embodiments, the light emitting unit **150** may include fewer or more light emitting segments. Additionally, although the illustrated lighting emitting unit **150** includes one light emitting segment corresponding to each actuator, in other embodiments, the light emitting unit **150** may include a different number of light emitting segments from the actuators, and/or the user interface **145** may include fewer or more actuators. Similar to the actuators, the illustrated light emitting segments **200, 205, 210, 215, 220** are arranged in a line that is parallel to the longitudinal axis A. In other embodiments, the light emitting segments **200, 205, 210, 215, 220** may be arranged in a line that is angled relative to the longitudinal axis A.

[0041] In some embodiments, each light emitting segment **200, 205, 210, 215, 220** includes one light emitting diode (LED). Additionally, or alternatively, in some embodiments, each light

emitting segment **200, 205, 210, 215, 220** includes a plurality of LEDs. In some embodiments, each light emitting segment **200, 205, 210, 215, 220** may be comprised of color-changing LEDs. For example, the LEDs may communicate data to a user using different colors, such as red, blue, green, etc. In some embodiments, the segments **200, 205, 210, 215, 220** are arranged sequentially on the housing **105** without interruption (i.e., a portion of the housing **105** does not extend between adjacent segments **200, 205, 210, 215, 220**). In other words, the segments **200, 205, 210, 215, 220** are arranged in a row, a continuous line, a strip, or an uninterrupted line on the housing **105**.

[0042] FIG. 4 is a communication network **300** for the audio output device **100**, an external device **305**, and a second audio output device **400**, according to some embodiments. As noted above, the audio output device **100** may communicate with an external device **305**. In such embodiments, the audio output device **100** may include, for example, a wireless communication controller (e.g., the Bluetooth® controller **515**) described above, which includes a transceiver to communicate with the external device **305** via, for example, a short-range communication protocol, such as Bluetooth®. The external device **305** may include a short-range transceiver, such as Bluetooth® controller **615** (FIG. 6), to communicate with the audio output device **100**, and may also include a long-range transceiver, such as communication interface **620** (FIG. 6), to communicate with a server (not shown). In some embodiments, a wired connection (via, for example, a USB cable) is provided between the external device **305** and the audio output device **100** to enable direct communication between the external device **305** and the audio output device **100**.

[0043] The external device **305** may include, for example, a smart telephone, a tablet computer, a cellular phone, a laptop computer, a smart watch, and any other communication device that is external to the audio output device **100**. The external device **305** may communicate with the audio output device **100** and may provide content (e.g., audio) to be played out of the speaker **130** of the audio output device **100**. For example, the audio may be sent as a first data packet from the external device **305** to the speaker **130**.

[0044] The audio output device **100** may be daisy chained to a second audio output device **400** to extend the audible range of the content being played from the audio output devices **100, 400** or to create a surround sound experience using the audio output devices **100, 400**. In some embodiments, the audio output device **100** is the parent device and the second audio output device **400** is the child device. The content may be played simultaneously from both audio output devices **100, 400**. Additional audio output devices may be added to the daisy chain. For example, the audio output device **100** may be the parent device to 2-5 child devices. The second audio output device **400** may be identical to the audio output device **100** and may include the same user interface, light emitting unit, and speaker, among other similar features.

[0045] When the second audio output device **400** moves out of range of the audio output device **100**, content playing over the speakers of the audio output device **100**, such as speaker **130**, and the second audio output device **400** is paused until the second audio output device **400** moves back in range. When the second audio output device **400** moves out of range of the audio output device **100** for 30 minutes or more, the second audio output device **400** is turned OFF. The audio output device **100** has control over the second audio output device **400**, as well as any additional daisy chained audio output devices. The volume of each daisy chained audio output device may be controlled at the audio output device **100** (e.g., via the volume actuators **175, 180**). The play/pause of each daisy chained audio output device may be controlled at the audio output device **100** (e.g., via the play actuator **170**). The forward and back control of each daisy chained audio output device may be controlled at the audio output device **100**. Child audio output devices, such as the second audio output device **400** may independently adjust the volume of their speakers (the volume delta will be maintained when the volume is then adjusted by the parent audio output device), control the mute/unmute of the second audio output device **400** when a play/pause actuator of the second audio output device **400** is actuated, and be removed from the daisy chain via actuation of a daisy chain actuator.

[0046] A controller **500** of the audio output device **100** is illustrated in FIG. 5. In some embodiments, the controller **500** is the same in each audio output device. The controller **500** is electrically and/or communicatively connected to a variety of modules or components of the audio output device **100**. For example, the illustrated controller **500** is connected to one or more sensors **505** (which may include, for example, current sensors, voltage sensors, temperature sensor, timers, etc., or a combination thereof), one or more indicators **510**, the battery pack **120**, a Bluetooth® controller **515**, the power actuator **160**, the volume actuators **175**, **180**, and the speakers **130**.

[0047] The controller **500** includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components and modules within the controller **500** and/or heated garment. For example, the controller **500** includes, among other things, a processing unit **520** (e.g., a microprocessor, an electronic processor, an electronic controller, a microcontroller, or another suitable programmable device), a memory **525**, input units **530**, and output units **535**. The processing unit **520** includes, among other things, a control unit **545**, an arithmetic logic unit (“ALU”) **550**, and a plurality of registers **555** (shown as a group of registers in FIG. 5), and is implemented using one or more computer architecture (e.g., a modified Harvard architecture, a von Neumann architecture, etc.). The processing unit **520**, the memory **525**, the input units **530**, and the output units **535**, as well as the various modules connected to the controller **500** are connected by one or more control and/or data buses (e.g., common bus **540**). The control and/or data buses are shown generally in FIG. 5 for illustrative purposes. The use of one or more control and/or data buses for the interconnection between and communication among the various modules and components would be known to a person skilled in the art in view of the embodiments described herein.

[0048] The memory **525** is a non-transitory computer readable medium and includes, for example, a program storage area and a data storage area. The program storage area and the data storage area can include combinations of different types of memory, such as a ROM, a RAM (e.g., DRAM, SDRAM, etc.), EEPROM, flash memory, a hard disk, an SD card, or other suitable magnetic, optical, physical, or electronic memory devices. The processing unit **520** is connected to the memory **525** and executes software instructions that are capable of being stored in a RAM portion of the memory **525** (e.g., during execution), a ROM portion of the memory **525** (e.g., on a generally permanent basis), or another non-transitory computer readable medium, such as another memory or a disc. Software included in the implementation of the audio output device **100** can be stored in the memory **525** of the controller **500**. The software includes, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The controller **500** is configured to retrieve from the memory **525** and execute, among other things, instructions related to the control processes and methods described herein. In other embodiments, the controller **500** includes additional, fewer, or different components.

[0049] The indicators **510** receive control signals from the controller **500** to turn ON and OFF or otherwise convey information based on different inputs to the controller **500**. In some embodiments, the indicators **510** may include the light emitting unit **150** (including light emitting segments **200**, **205**, **210**, **215**, **220**) and the fuel gauge **165**. For example, the light emitting segments **200**, **205**, **210**, **215**, **220** may illuminate when the audio output device **100** is powered ON, when the audio output device **100** enters Bluetooth® pairing mode, when the audio output device **100** pairs with a device via Bluetooth®, when the audio output device **100** is powered OFF, when the audio output device **100** loses Bluetooth® connection, when the audio output device **100** is daisy chained with a second audio output device **400**, when the play actuator **170** is actuated, when the volume actuators **175**, **180** are actuated, and the like. The fuel gauge **165** may display the state of charge of the battery pack **120**. For example, the fuel gauge **165** may display the state of charge continuously while the audio output device **100** is ON. As another example, the fuel gauge **165** may temporarily or briefly display the state of charge upon powering ON of the audio output

device **100**. After a short period of time (e.g., 3 seconds), the fuel gauge **165** may turn off (i.e., stop displaying the state of charge).

[0050] The indicators **510** include, for example, one or more light-emitting diodes (LEDs), a display screen (e.g., an LCD display), or a combination thereof. The display/indicator(s) **510** may also include additional elements to convey information to a user through one or more audible outputs, tactile outputs (e.g., a speaker), or a combination thereof. The display/indicator(s) **510** may also be referred to as an output device configured to provide an output to a user.

[0051] The controller **500** may receive an input from the power actuator **160** to turn ON/OFF the audio output device **100**. The controller **500** may further receive an input from the volume actuators **175, 180** to increase/decrease the volume of the content playing over the speakers **130**. The controller **500** may also receive input from the play actuator **170** to play, pause, skip forward, or skip backward the content playing over the speaker **130**. Additionally, the controller **500** may receive input from the daisy chain actuator **190**.

[0052] The Bluetooth® controller **515** enables the audio output device **100** to communicate with other Bluetooth® enabled devices, such as the Bluetooth® controller **615** of the external device **305** or a Bluetooth® controller **615** of a second audio output device **400**, when the audio output device **100** is in range of the other devices. In some embodiments, the audio output device **100** may communicate with other devices over a network (not shown). The Bluetooth® controller **515** includes a transceiver to send and receive data to other devices. In some embodiments, the Bluetooth® controller **515** pairs with other devices when the Bluetooth® actuator **180** is actuated by a user of the audio output device **100**.

[0053] FIG. **6** is a block diagram of the external device **305**. In some embodiments, the external device **305** may include one or more processors **600**, a memory **605**, a user interface **610**, the Bluetooth® controller **615**, and the communication interface **620**. A memory bus may be used for communication between the processor **600** and the memory **605**.

[0054] The processor **600** may be, for example, an electronic processor, a microprocessor (μ P), a microcontroller (μ C), a digital signal processor (DSP), or a combination thereof. The processor **600** may include one or more levels of caching, such as a level cache memory, a processor core, and registers. The processor core may include an arithmetic logic unit (ALU), a floating point unit (FPU), a digital signal processing core (DSP Core), or any combination thereof. A memory controller may also be used with the processor **600**, or in some implementations, the memory controller may be an internal part of the processor **600**. In some embodiments, the memory **605** may be an application for playing content. For example, the processor **600** may execute an audio application (app) stored in the memory **605** based on input received at the user interface **610** which sends content (e.g., audio) to the audio output device **100**. Depending on the desired configuration, the memory **605** include, for example, volatile memory (such as RAM), non-volatile memory (such as ROM, flash memory, etc.), or a combination thereof.

[0055] The user interface **610** may include a hardware screen that may be communicatively coupled to the external device **305**. The user interface **610** may include a touch sensitive device that detects gestures such as a touch action. The user interface **610** may also provide feedback in response to detected gestures (or other forms of input).

[0056] The Bluetooth® controller **615** enables the external device **305** to communicate with other Bluetooth® enabled devices, such as the Bluetooth® controller **515** of the audio output device **100**, when the external device **305** is in range of the other devices. In some embodiments, the external device **305** may communicate with other devices over a network (not shown) via the communication interface **620**. For example, the communication interface **620** may include a transceiver. As previously noted, the devices described herein are not limited to communicating via the Bluetooth® protocol and may use other forms and protocols of wireless communication.

[0057] The external device **305** may have additional features or functionality, and additional interfaces to facilitate communications between the external device **305** and other devices (e.g.,

audio output device **100**) or networks. The external device **305** may also have additional internal components not illustrated in FIG. **6**.

[0058] FIGS. **7A-16** depict exemplary methods or operations of the audio output device **100**. The light emitting unit **150** is configured to output or display different light sequences based on or in response to operation of the audio output device **100**. For example, the light emitting unit **150** may output a light sequence in response to actuation of an actuator **160, 170, 175, 180, 185, 190**. The light sequences may be different patterns of illumination of the light emitting segments **200, 205, 210, 215, 220**. For example, the light emitting segments **200, 205, 210, 215, 220** may illuminate at different times. In addition, the light emitting segments **200, 205, 210, 215, 220** may illuminate at different brightnesses or intensities. Furthermore, the light emitting segments **200, 205, 210, 215, 220** may illuminate in different colors. As such, the light emitting unit **150** may output different patterns of wave effects, ripple effects, alternating patterns, flashing patterns, asymmetric patterns, irregular patterns, and the like to indicate a current operation or state of the audio output device **100**. For some operations, the light emitting unit **150** may output a first light sequence at a start of an operation to indicate the operation is being initiated, and may output a second light sequence at an end of the operation to indicate the operation is complete.

[0059] In addition to outputting light sequences via the light emitting unit **150**, the audio output device **100** may also output tones associated with an operation or actuation of an actuator **160, 170, 175, 180, 185, 190** through the speaker **130**. In some embodiments, the tone may be synched with the light sequence. That is, the tone may be played with or at least start with the light sequence. In other embodiments, the tone may be played before the light sequence begins.

[0060] FIGS. **7A** and **7B** are a method **700** of powering ON the audio output device **100** and pairing the audio output device **100** with an external device, such as external device **305**, upon powering ON, according to some embodiments. Although the illustrated method **700** includes specific steps, not all of the steps need to be performed or need to be performed in the order presented. In some embodiments, the method **700** is executed by the audio output device controller **500**. The values of various thresholds or ranges described with respect to method **700** are provided as non-limiting examples and other values for such parameters are possible and may vary based on the type or condition of the audio output device **100**, the type or condition of the external device **305**, other factors, or a combination thereof.

[0061] The method **700** includes the controller **500** determining that the power actuator **160** of the audio output device **100** is actuated for a predetermined amount of time (step **705**). In some embodiments, the predetermined amount of time is a first predetermined amount of time. For example, the first predetermined amount of time may be any time less than 750 milliseconds (ms). In step **708**, the controller **500** turns ON the audio output device **100**. In some embodiments, turning ON the audio output device **100** includes allowing the flow of power from the battery pack **120** to the audio output device components coupled to the controller **500**. In step **710**, the controller **500** outputs a first tone to the speaker **130**. In some embodiments, the first tone is a certain pitch and frequency. For example, the first tone may be a “ramp-up” tone that increases exponentially in pitch or frequency over the course of one second. In some embodiments, the first tone is volume independent.

[0062] In step **715**, the controller **500** turns ON the light emitting unit **150**. In some embodiments, each segment **200, 205, 210, 215, 220** illuminates. In step **720**, the controller **500** provides a first light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the first light sequence. In some embodiments, the first light sequence includes illuminating the third segment **210**, implementing a 100 ms delay, illuminating the second segment **205** and the fourth segment **215**, implementing another 100 ms delay, and illuminating the first segment **200** and the fifth segment **220**. For example, the first light sequence may be a “ripple” light sequence.

[0063] In step **725**, the controller **500** turns ON the fuel gauge **165** of the audio output device **100**.

In step **730**, the controller provides a first state of charge signal of the battery pack **120** to be indicated by the fuel gauge **165**. For example, a first LED of four LEDs comprising the fuel gauge **165** may be illuminated to indicate or display a low state of charge of the battery pack **120**. Alternatively, all four LEDs comprising the fuel gauge **165** may be illuminated to indicate or display a full or high state of charge of the battery pack **120**. The method continues to FIG. 7B. In some embodiments, when the state of charge of the battery pack **120** is low, a low battery tone plays every 60 seconds until the battery pack **120** dies, the audio output device **100** is powered OFF, or a new battery pack is inserted. For example, the low battery tone may be two distinct sounds in descending pitch/frequency.

[0064] In step **735**, the controller **500** enters into a Bluetooth® pairing mode. In some embodiments, the Bluetooth® controller **515** of the audio output device **100** broadcasts a unique key (e.g., a UUID) at a set interval when in pairing mode. In step **740**, the controller **500** outputs a second tone to the speaker **130**. In some embodiments, the second tone is a certain pitch and frequency. For example, the second tone may be three distinct sounds each a first length in time in ascending pitch or frequency that are repeated throughout the course of the pairing mode. In some embodiments, the second tone is volume independent. In step **745**, the controller oscillates illumination of the Bluetooth® actuator **185**. In some embodiments, an LED of the Bluetooth® actuator **185** flashes slowly at a rate of 1.0 Hertz (Hz) while the audio output device **100** searches for a Bluetooth® capable device, such as external device **305**.

[0065] In step **750**, the controller **500** provides a second light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the second light sequence. In some embodiments, the second light sequence includes turning OFF each segment **200**, **205**, **210**, **215**, **220**, implementing a 100 ms delay, illuminate the third segment **210**, implementing another 100 ms delay, illuminating the second segment **205** and the fourth segment **215**, dimming the third segment **210** to a 30% illumination level (i.e., brightness), implementing another 100 ms delay, turning OFF the third segment **210**, dimming the second segment **205** and the fourth segment **215** to the 30% illumination level, illuminating the first segment **200** and the fifth segment **220**, implementing another 100 ms delay, turning OFF the second segment **205** and the fourth segment **215**, dimming the first segment **200** and the fifth segment **220** to the 30% illumination level, implementing another 100 ms delay, turning OFF the first segment **200** and the fifth segment **220**, implementing a 2200 ms delay, and then repeating the second light sequence starting at illuminating the third segment **210** again until pairing is complete or times out.

[0066] In decision step **755**, the controller **500** determines whether the audio output device **100** is paired with an external device, such as external device **305**. In some embodiments, the Bluetooth® controller **515** receives a receipt from the Bluetooth® controller **615** of the external device **305** to determine that the external device **305** is paired with the audio output device **100**. When the controller **500** determines that the audio output device **100** is paired with the external device **305**, the method **700** proceeds to step **760**. When the controller **500** determines that the audio output device **100** is not paired with the external device **305**, the method **700** proceeds to step **770**. In some embodiments, the controller **500** tries to pair the audio output device **100** with the external device **305** for three minutes before determining that the audio output device **100** is not paired with the external device **305**.

[0067] In step **760** (PAIRED at decision step **755**), the controller **500** outputs a third tone to the speaker **130**. In some embodiments, the third tone is a certain pitch and frequency. For example, the third tone may be two distinct sounds each a second length in time in ascending pitch or frequency. In some embodiments, the third tone is volume independent. In step **765**, the controller **500** outputs a third light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the third light sequence. In some embodiments, the third light sequence includes turning OFF each segment **200**, **205**, **210**, **215**, **220**,

illuminating the first segment **200** and the fifth segment **220**, implementing a 100 ms delay, illuminating the second segment **205** and the fourth segment **215**, implementing another 100 ms delay, and illuminating the third segment **210**.

[0068] In step **770** (NOT PAIRED at decision step **755**), the controller **500** provides a fourth light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the fourth light sequence. In some embodiments, the fourth light sequence includes illuminating each segment **200**, **205**, **210**, **215**, **220** and oscillating illumination of the segments at a rate of 0.5 Hz for a predetermined amount of time before illuminating each segment **200**, **205**, **210**, **215**, **220** solid. In step **775**, the controller **500** determines that the Bluetooth® actuator has been actuated. The method **700** then begins again at step **735** to reinitiate a pairing mode of the audio output device **100**. In some embodiments, the controller **500** may pair the audio output device **100** with a new external device (different from the paired external device **305**) when the Bluetooth® actuator is actuated for more than 750 ms. The controller **500** then repeats the pairing mode of the audio output device **100**.

[0069] FIG. **8** is a method **800** of powering OFF the audio output device **100**, according to some embodiments. Although the illustrated method **800** includes specific steps, not all of the steps need to be performed or need to be performed in the order presented. In some embodiments, the method **800** is executed by the audio output device controller **500**. The values of various thresholds or ranges described with respect to method **800** are provided as non-limiting examples and other values for such parameters are possible and may vary based on the type or condition of the audio output device **100**, the type or condition of the external device **305**, other factors, or a combination thereof.

[0070] The method **800** includes the controller **500** determining that the power actuator **160** is actuated for a predetermined amount of time or that the audio output device **100** has not been used for a second predetermined amount of time (step **805**). In some embodiments, the predetermined amount of time is a first predetermined amount of time. For example, the first predetermined amount of time may be any time less than 750 milliseconds (ms). In some embodiments, the second predetermined amount of time is 30 minutes. In some embodiments, not in use means that the audio output device **100** has not played any content over the speaker **130**.

[0071] In step **810**, the controller **500** outputs a fourth tone to the speaker **130**. In some embodiments, the fourth tone is a certain pitch and frequency. For example, the fourth tone may be a “ramp-down” tone that decreases exponentially in pitch or frequency over the course of one second. In some embodiments, the fourth tone is volume independent. In step **815**, the controller **500** provides a fifth light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the fifth light sequence. In some embodiments, the fifth light sequence includes turning OFF the first segment **200** and the fifth segment **220**, implementing a 100 ms delay, turning OFF the second segment **205** and the fourth segment **215**, implementing another 100 ms delay, and turning OFF the third segment **210**. For example, the fifth light sequence may be a reverse “ripple” light sequence that completely turns OFF the light emitting unit **150**.

[0072] In step **820**, the controller **500** turns OFF the fuel gauge **165**. In step **825**, the controller **500** turns OFF the audio output device **100**. In some embodiments, the controller **500** cuts off the flow of power from the battery pack **120** to the audio output device components coupled to the controller **500** to turn OFF the audio output device **100**.

[0073] FIGS. **9A** and **9B** are a method **900** of pairing the audio output device **100** with an external device, such as external device **305**, upon powering ON, according to some embodiments. Although the illustrated method **900** includes specific steps, not all of the steps need to be performed or need to be performed in the order presented. In some embodiments, the method **900** is executed by the audio output device controller **500**. The values of various thresholds or ranges described with respect to method **900** are provided as non-limiting examples and other values for

such parameters are possible and may vary based on the type or condition of the audio output device **100**, the type or condition of the external device **305**, other factors, or a combination thereof. [0074] In contrast to the method **800**, the illustrated method **900** may be used to pair the audio output device **100** with an external device the first time the audio output device **100** is used or to pair the audio output device **100** with a different external device. The method **900** includes the controller **500** determining that the Bluetooth® actuator has been actuated (step **905**). In some embodiments, the predetermined amount of time is a first predetermined amount of time. For example, the first predetermined amount of time may be any time less than 750 milliseconds (ms). In step **910**, the controller **500** enters into a Bluetooth® pairing mode. In some embodiments, the Bluetooth® controller **515** of the audio output device **100** broadcasts a unique key (e.g., a UUID) at a set interval when in pairing mode. In step **915**, the controller **500** outputs a fifth tone to the speaker **130**. In some embodiments, the fifth tone is a certain pitch and frequency. For example, the fifth tone may be three distinct sounds each a first length in time in ascending pitch or frequency that are repeated throughout the course of the pairing mode every 5 seconds. In some embodiments, the fifth tone is volume independent. In step **920**, the controller illuminates the Bluetooth® actuator **185**. In some embodiments, an LED of the Bluetooth® actuator **185** flashes slowly at a rate of 1.0 Hertz (Hz) while the audio output device **100** searches for a Bluetooth® capable device, such as external device **305**.

[0075] In step **925**, the controller **500** provides a sixth light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the sixth light sequence. In some embodiments, the sixth light sequence includes turning OFF each segment **200**, **205**, **210**, **215**, **220**, implementing a 100 ms delay, illuminate the third segment **210**, implementing another 100 ms delay, illuminating the second segment **205** and the fourth segment **215**, dimming the third segment **210** to a 30% illumination level (i.e., brightness), implementing another 100 ms delay, turning OFF the third segment **210**, dimming the second segment **205** and the fourth segment **215** to the 30% illumination level, illuminating the first segment **200** and the fifth segment **220**, implementing another 100 ms delay, turning OFF the second segment **205** and the fourth segment **215**, dimming the first segment **200** and the fifth segment **220** to the 30% illumination level, implementing another 100 ms delay, turning OFF the first segment **200** and the fifth segment **220**, implementing a 2200 ms delay, and then repeating the second light sequence starting at illuminating the third segment **210** again until pairing is complete or times out.

[0076] In decision step **930**, the controller **500** determines whether the audio output device **100** is paired with an external device, such as external device **305**. In some embodiments, the Bluetooth® controller **515** receives a receipt from the Bluetooth® controller **615** of the external device **305** to determine that the external device **305** is paired with the audio output device **100**. When the controller **500** determines that the audio output device **100** is paired with the external device **305**, the method **900** proceeds to step **935**. When the controller **500** determines that the audio output device **100** is not paired with the external device **305**, the method **900** proceeds to step **950**. In some embodiments, the controller **500** tries to pair the audio output device **100** with the external device **305** for three minutes before determining that the audio output device **100** is not paired with the external device **305**.

[0077] In step **935** (PAIRED at decision step **755**), the controller **500** outputs the third tone to the speaker **130**. In some embodiments, the third tone is a certain pitch and frequency. For example, the third tone may be two distinct sounds each a second length in time in ascending pitch or frequency. In some embodiments, the third tone is volume independent. In step **940**, the controller **500** outputs the third light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the third light sequence. In some embodiments, the third light sequence includes turning OFF each segment **200**, **205**, **210**, **215**, **220**, illuminating the first segment **200** and the fifth segment **220**, implementing a 100 ms delay,

illuminating the second segment **205** and the fourth segment **215**, implementing another 100 ms delay, and illuminating the third segment **210**. In step **945**, the controller **500** illuminates the Bluetooth® actuator **185**. In some embodiments, the LED of the Bluetooth® actuator illuminates solid. The method **900** continues to FIG. **9B**.

[0078] In step **950** (NOT PAIRED at decision step **930**), the controller **500** remains paired with the last connected external device. In step **955**, the controller **500** determines that the Bluetooth® actuator **185** has been actuated. The method **900** then begins again at step **910** to reinitiate a pairing mode of the audio output device **100**.

[0079] In decision step **960**, the controller **500** determines whether the external device **305** is playing content. When the controller **500** determines that the external device **305** is playing content (e.g., on an audio app), the method **900** proceeds to step **965**. When the controller **500** determines that the external device **305** is not playing content, the method **900** proceeds to step **975**.

[0080] In step **965** (PLAYING at decision step **960**), the controller **500** receives content from the external device **305**. In step **970**, the controller **500** outputs the content over the speaker **130**. In some embodiments, the content is played at the volume that is set on the external device **305**. In step **975** (NOT PLAYING at decision step **960**), the controller **500** controls the speaker **130** to remain OFF.

[0081] FIG. **10** is a method **1000** of determining that the Bluetooth® connection with the external device **305** is lost, according to some embodiments. Although the illustrated method **1000** includes specific steps, not all of the steps need to be performed or need to be performed in the order presented. In some embodiments, the method **1000** is executed by the audio output device controller **500**. The values of various thresholds or ranges described with respect to method **1000** are provided as non-limiting examples and other values for such parameters are possible and may vary based on the type or condition of the audio output device **100**, the type or condition of the external device **305**, other factors, or a combination thereof.

[0082] The method **1000** includes the controller **500** determining that the Bluetooth® connection with the external device **305** is lost (step **1005**). In some embodiments, the Bluetooth® connection is lost when the audio output device **100** moves out of range of the external device **305**. In step **1010**, the controller **500** outputs a sixth tone to the speaker **130**. In some embodiments, the sixth tone is a certain pitch and frequency. For example, the sixth tone may be two distinct sounds each a second length in time in descending pitch or frequency that is played every 5 seconds until a Bluetooth® connection is found. In some embodiments, the sixth tone is volume independent.

[0083] In step **1015**, controller **500** illuminates the Bluetooth® actuator **185**. In some embodiments, and LED of the Bluetooth® actuator **185** flashes slowly at a rate of 1.0 Hertz (Hz) while the audio output device **100** restores the Bluetooth® connection. In step **1020**, the controller **500** provides a seventh light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the seventh light sequence. In some embodiments, the seventh light sequence includes illuminating each segment **200**, **205**, **210**, **215**, **220** and oscillating illumination of the segments at a rate of 0.5 Hz for a predetermined amount of time before illuminating each segment **200**, **205**, **210**, **215**, **220** solid.

[0084] In decision step **1025**, the controller **500** determines whether the audio output device **100** is paired with an external device, such as external device **305**. In some embodiments, the Bluetooth® controller **515** receives a receipt from the Bluetooth® controller **615** of the external device **305** to determine that the external device **305** is paired with the audio output device **100**. When the controller **500** determines that the audio output device **100** is paired with the external device **305**, the method **1000** proceeds to step **1030**. When the controller **500** determines that the audio output device **100** is not paired with the external device **305**, the method **1000** proceeds to step **1040**. In some embodiments, the controller **500** tries to pair the audio output device **100** with the external device **305** for three minutes before determining that the audio output device **100** is not paired with the external device **305**.

[0085] In step **1030** (PAIRED at decision step **1025**), the controller **500** outputs the third tone to the speaker **130**. In some embodiments, the third tone is a certain pitch and frequency. For example, the third tone may be two distinct sounds each a second length in time in ascending pitch or frequency. In step **1035**, the controller **500** outputs the third light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the third light sequence. In some embodiments, the third light sequence includes turning OFF each segment **200**, **205**, **210**, **215**, **220**, illuminating the first segment **200** and the fifth segment **220**, implementing a 100 ms delay, illuminating the second segment **205** and the fourth segment **215**, implementing another 100 ms delay, and illuminating the third segment **210**.

[0086] In step **1040** (NOT PAIRED at decision step **1025**), the controller **500** turns OFF the Bluetooth® actuator **185**. For example, the controller **500** turns OFF the LED of the Bluetooth® actuator **185**. In step **1045**, the controller **500** determines that the Bluetooth® actuator **185** is actuated to reinitiate Bluetooth® pairing.

[0087] FIG. **11** is a method **1100** of daisy chaining the audio output device **100** with a second audio output device, such as audio output device **400**, according to some embodiments. Although the illustrated method **1100** includes specific steps, not all of the steps need to be performed or need to be performed in the order presented. In some embodiments, the method **1100** is executed by the audio output device controller **500**. The values of various thresholds or ranges described with respect to method **1100** are provided as non-limiting examples and other values for such parameters are possible and may vary based on the type or condition of the audio output device **100**, the type or condition of the external device **305**, other factors, or a combination thereof.

[0088] The method **1100** includes the controller **500** determining that a daisy chain actuator of a first audio output device, such as audio output device **100**, is actuated for a predetermined amount of time (step **1105**). In some embodiments, the predetermined amount of time is a first predetermined amount of time. For example, the first predetermined amount of time may be any time less than 750 milliseconds (ms). In step **1110**, the controller **500** outputs a seventh tone to the speaker **130**. In some embodiments, the seventh tone is a certain pitch and frequency. For example, the seventh tone may be three distinct sounds each a first length in time in ascending pitch or frequency that are repeated throughout the course of the daisy chaining mode. In some embodiments, the seventh tone is a higher pitch than the first tone. In some embodiments, the seventh tone is volume independent. In step **1115**, the controller **500** illuminates the daisy chain actuator **190**. In some embodiments, an LED of the daisy chain actuator **190** illuminates solid.

[0089] In step **1120**, the controller **500** provides an eighth light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the eighth light sequence. In some embodiments, the eighth light sequence includes turning OFF each segment **200**, **205**, **210**, **215**, **220**, illuminating the first segment **200**, implementing a 100 ms delay, dimming the first segment **200** to a 60% illumination level (i.e., brightness), illuminating the second segment **205**, implementing another 100 ms delay, further dimming the first segment **200** to a 30% illumination level, dimming the second segment **205** to the 60% illumination level, illuminating the third segment **210**, implementing another 100 ms delay, turning OFF the first segment **200**, dimming the second segment **205** to the 30% illumination level, dimming the third segment **210** to the 60% illumination level, turning ON the fourth segment **215**, implementing another 100 ms delay, turning OFF the second segment **205**, dimming the third segment **210** to the 30% illumination level, dimming the fourth segment **215** to the 60% illumination level, turning ON the fifth segment **220**, implementing another 100 ms delay, turning OFF the third segment **210**, dimming the fourth segment **215** to the 30% illumination level, implementing another 100 ms delay, turning OFF the fourth segment **215**, implementing another 100 ms delay, illuminating the fourth segment **215**, dimming the fifth segment **220** to the 60% illumination level, implementing another 100 ms delay, illuminating the third segment **210**, dimming the fourth segment **215** to the 60% illumination level, dimming the fifth segment **220** to the 30% illumination level, and

implementing another 100 ms delay, turning OFF the fifth segment **220**, dimming the fourth segment **215** to the 30% illumination level, dimming the third segment **210** to the 60% illumination level, illuminating the second segment **205**, implementing another 100 ms delay, turning OFF the fourth segment **215**, dimming the third segment **210** to the 30% illumination level, dimming the second segment **205** to the 60% illumination level, illuminating the first segment **200**, implementing another 100 ms delay, turning OFF the third segment **210**, dimming the second segment **205** to the 30% illumination level, and implementing a 100 ms delay. The eighth light sequence repeats twice while the audio output device **100** daisy chains.

[0090] In decision step **1125**, the controller **500** determines whether the audio output device **100** is daisy chained with another audio output device, such as the second audio output device **400**. When the controller **500** determines that the audio output device **100** is daisy chained with the second audio output device **400**, the method **1100** proceeds to step **1130**. When the controller **500** determines that the audio output device **100** is not daisy chained with the second audio output device **400**, the method **1100** proceeds to step **1140**. In some embodiments, the controller **500** tries to daisy chain the audio output device **100** with the second audio output device **400** for three minutes before determining that the audio output device **100** is not daisy chained with the second audio output device **400**.

[0091] In step **1130** (DAISY CHAINED at decision step **1125**), the controller **500** outputs an eighth tone to the speaker **130**. In some embodiments, the eighth tone is a certain pitch and frequency. For example, the eighth tone may be two distinct sounds each a third length in time in ascending pitch or frequency. In some embodiments, the eighth tone is volume independent. In step **1135**, the controller **500** outputs a ninth light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the ninth light sequence. In some embodiments, the ninth light sequence includes illuminating the first segment **200**, implementing a 100 ms delay, illuminating the second segment **205**, implementing a 100 ms delay, illuminating the third segment **210**, implementing a 100 ms delay, illuminating the fourth segment **215**, implementing a 100 ms delay, and illuminating the fifth segment **210**.

[0092] In step **1140** (NOT DAISY CHAINED at decision step **1125**), the controller **500** turns OFF the daisy chain actuator **190** and the light emitting unit **150**. For example, the controller **500** turns OFF the LED of the daisy chain actuator **190** and each segment **200**, **205**, **210**, **215**, **220**. In step **1145**, the controller **500** determines that the daisy chain actuator **190** is actuated to reinitiate the daisy chain mode. The method **1100** then begins again at step **1110** to attempt to daisy chain the audio output device **100** with the second audio output device **400**.

[0093] FIG. **12** is a method **1200** of daisy chaining the second audio output device **400** with a third audio output device, according to some embodiments. Although the illustrated method **1200** includes specific steps, not all of the steps need to be performed or need to be performed in the order presented. In some embodiments, the method **1200** is executed by a controller of the second audio output device **400**. The controller of the second audio output device **400** is the same as the controller **500** of the audio output device **100**. The values of various thresholds or ranges described with respect to method **1200** are provided as non-limiting examples and other values for such parameters are possible and may vary based on the type or condition of the second audio output device **400**, the type or condition of the external device **305**, other factors, or a combination thereof.

[0094] The method **1200** includes the controller of a second audio output device determining that a daisy chain actuator of a second audio output device, such as second audio output device **400**, is actuated for a predetermined amount of time (step **1205**). In some embodiments, the predetermined amount of time is a first predetermined amount of time. For example, the first predetermined amount of time may be any time less than 750 milliseconds (ms). In step **1210**, the controller of the second audio output device **400** outputs a ninth tone to the speaker of the second audio output device **400**. In some embodiments, the ninth tone is a certain pitch and frequency. For example, the ninth tone may be three distinct sounds each a first length in time in ascending pitch or frequency

that are repeated throughout the course of the daisy chaining mode. In some embodiments, the ninth tone is a higher pitch than the first tone and substantially the same as the seventh tone. In some embodiments, the ninth tone is volume independent. In step **1215**, the controller of the second audio output device **400** illuminates a daisy chain actuator of the second audio output device **400**. In some embodiments, an LED of the daisy chain actuator of the second audio output device **400** flashes slowly at a rate of 1.0 Hertz (Hz) while the second audio output device **400** searches for a third audio output device.

[0095] In step **1220**, the controller of the second audio output device **400** provides a tenth light sequence to be emitted by a light emitting unit of the second audio output device **400**. In other words, the controller operates the light emitting unit of the second audio output device **400** according to the tenth light sequence. In some embodiments, the tenth light sequence is the same as the eighth light sequence. In some embodiments, the tenth light sequence includes illuminating the first segment, implementing a 100 ms delay, dimming the first segment to a 60% illumination level (i.e., brightness), illuminating the second segment, implementing another 100 ms delay, further dimming the first segment to a 30% illumination level, dimming the second segment to the 60% illumination level, illuminating the third segment, implementing another 100 ms delay, turning OFF the first segment, dimming the second segment to the 30% illumination level, dimming the third segment to the 60% illumination level, turning ON the fourth segment, implementing another 100 ms delay, turning OFF the second segment, dimming the third segment to the 30% illumination level, dimming the fourth segment to the 60% illumination level, turning ON the fifth segment, implementing another 100 ms delay, turning OFF the third segment, dimming the fourth segment to the 30% illumination level, implementing another 100 ms delay, turning OFF the fourth segment, implementing another 100 ms delay, illuminating the fourth segment, dimming the fifth segment to the 60% illumination level, implementing another 100 ms delay, illuminating the third segment, dimming the fourth segment to the 60% illumination level, dimming the fifth segment to the 30% illumination level, and implementing another 100 ms delay, turning OFF the fifth segment, dimming the fourth segment to the 30% illumination level, dimming the third segment to the 60% illumination level, illuminating the second segment, implementing another 100 ms delay, turning OFF the fourth segment, dimming the third segment to the 30% illumination level, dimming the second segment to the 60% illumination level, illuminating the first segment, implementing another 100 ms delay, turning OFF the third segment, dimming the second segment to the 30% illumination level, and implementing a 100 ms delay. The tenth light sequence repeats twice while the second audio output device **400** is in the daisy chain mode.

[0096] In step **1225**, the controller of the second audio output device **400** determines that the second audio output device **400** is daisy chained with a third audio output device. In some embodiments, the third audio output device is identical to the audio output device **100** and the second audio output device **400**.

[0097] FIG. **13** is a method **1300** of removing the second audio output device **400** from a daisy chain, according to some embodiments. Although the illustrated method **1300** includes specific steps, not all of the steps need to be performed or need to be performed in the order presented. In some embodiments, the method **1300** is executed by a controller of the second audio output device **400**. The controller of the second audio output device **400** is the same as the controller **500** of the audio output device **100**. The values of various thresholds or ranges described with respect to method **1300** are provided as non-limiting examples and other values for such parameters are possible and may vary based on the type or condition of the second audio output device **400**, the type or condition of the external device **305**, other factors, or a combination thereof.

[0098] The method **1300** includes the controller of the second audio output device **400** determining that the daisy chain actuator of the second audio output device **400** is actuated for a predetermined amount of time (step **1305**). In some embodiments, the predetermined amount of time is a first predetermined amount of time. For example, the first predetermined amount of time may be any

time less than 750 milliseconds (ms). In step **1310**, the controller of the second audio output device **400** outputs a tenth tone to the speaker of the second audio output device **400**. In some embodiments, the tenth tone is a certain pitch and frequency. For example, the tenth tone may be two distinct sounds each a second length in time in descending pitch or frequency. In some embodiments, the tenth tone is volume independent. In step **1315**, the controller of the second audio output device **400** turns OFF the daisy chain actuator of the second audio output device **400**. [0099] In step **1320**, the controller of the second audio output device **400** provides an eleventh light sequence to be emitted by a light emitting unit of the second audio output device **400**. In other words, the controller operates the light emitting unit of the second audio output device **400** according to the eleventh light sequence. In some embodiments, the eleventh light sequence may be similar to one of the light sequences described above. In step **1325**, the controller of the second audio output device **400** determines that the second audio output device **400** is removed from the daisy chain and is an independent audio output device.

[0100] FIG. **14** is a method **1400** of playback control on the audio output device **100**, according to some embodiments. Although the illustrated method **1400** includes specific steps, not all of the steps need to be performed or need to be performed in the order presented. In some embodiments, the method **1400** is executed by the audio output device controller **500**. The values of various thresholds or ranges described with respect to method **1400** are provided as non-limiting examples and other values for such parameters are possible and may vary based on the type or condition of the audio output device **100**, the type or condition of the external device **305**, other factors, or a combination thereof.

[0101] The method **1400** includes the controller **500** determining that a first actuator is actuated. In some embodiments, the first actuator is the play actuator **170** (step **1405**). In some embodiments, play actuator **170** is actuated for a first predetermined amount of time when content is not playing over the speaker **130** of the audio output device **100** to initiate “playing” the content over the speaker. For example, the first predetermined amount of time may be any time less than 750 milliseconds (ms). Alternatively, or additionally, in some embodiments, the play actuator **170** is actuated for the first predetermined amount of time when content is playing over the speaker **130** of the audio output device **100** to pause the content over the speaker. Alternatively, or additionally, in some embodiments, the play actuator **170** is actuated twice to advance the content (e.g., skip ahead to play a next audio file). Alternatively, or additionally, in some embodiments, the play actuator **170** is actuated three times to make the content go backwards (e.g., skip backwards to play a last audio file). Alternatively, or additionally, in some embodiments, the play actuator **170** is actuated for a second predetermined amount of time to enter into an audio equalization mode. For example, the second predetermined amount of time may be any time greater than 1 second. In some embodiments, the controller **500** outputs a bass tone when the play actuator **170** is actuated for a second predetermined amount of time to enter into an audio equalization mode. In some embodiments, once in the audio equalization mode, the controller **500** may determine that the volume increase actuator **175** is actuated to increase the bass level or that the volume decrease actuator **180** is actuated to decrease the bass level. In some embodiments, the controller **500** determines that the play actuator **170** is once again actuated to set the bass level and to advance into adjusting a treble level. In some embodiments, the controller **500** outputs a treble tone when the play actuator **170** is once again actuated to set the bass level and to advance into adjusting a treble level. In some embodiments, once in the treble mode, the controller **500** may determine that the volume increase actuator **175** is actuated to increase the treble level or that the volume decrease actuator **180** is actuated to decrease the treble level. In some embodiments, the controller **500** determines that the play actuator **170** is once again actuated to set the treble level and to exit the audio equalization mode.

[0102] In step **1410**, the controller **500** performs the determined playback control. For example, based on the actuation of the play actuator **170** as described above, the controller performs a

specific control (e.g., play, pause, forward, backward, bass level, treble level). In step **1415**, the controller **500** provides a twelfth light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the twelfth light sequence. The twelfth light sequence is dependent on the playback control performed and is shown in Table 1, below.

TABLE-US-00001 TABLE 1 Twelfth Light Sequence Playback Control Twelfth Light Sequence
Play/Pause Each segment 200, 205, 210, 215, 220 flashes at a rate of 0.5 Hz Forward Each segment 200, 205, 210, 215, 220 turned OFF, 100 ms delay implemented, fifth segment 220 illuminated, another 100 ms delay implemented, fourth segment 215 illuminated, another 100 ms delay implemented, third segment 210 illuminated, another 100 ms delay implemented, second segment 205 illuminated, another 100 ms delay implemented, first segment 200 illuminated, each segment 200, 205, 210, 215, 220 flashes at a rate of 0.5 Hz Backward Each segment 200, 205, 210, 215, 220 turned OFF, 100 ms delay implemented, first segment 200 illuminated, another 100 ms delay implemented, second segment 205 illuminated, another 100 ms delay implemented, third segment 210 illuminated, another 100 ms delay implemented, fourth segment 215 illuminated, another 100 ms delay implemented, fifth segment 220 illuminated, each segment 200, 205, 210, 215, 220 flashes at a rate of 0.5 Hz Bass Level Each segment 200, 205, 210, 215, 220 flashes Increase & once at a rate of 0.5 Hz, third segment 210 Treble Level illuminated, 100 ms delay implemented, second Increase segment 205 illuminated, 100 ms delay implemented, first segment 200 illuminated, 100 ms delay implemented, third segment 210 dimmed to 40%, 100 ms delay implemented, second segment 205 dimmed to 40%, 100 ms delay implemented, first segment 200 dimmed to 40%, each segment 200, 205, 210, 215, 220 flash once at a rate of 0.5 Hz, Bass Level Each segment 200, 205, 210, 215, 220 flashes Decrease & once at a rate of 0.5 Hz, third segment 210 Treble Level illuminated, 100 ms delay implemented, fourth Decrease segment 215 illuminated, 100 ms delay implemented, fifth segment 220 illuminated, 100 ms delay implemented, third segment 210 dimmed to 40%, 100 ms delay implemented, fourth segment 215 dimmed to 40%, 100 ms delay implemented, fifth segment 220 dimmed to 40%, each segment 200, 205, 210, 215, 220 flash once at a rate of 0.5 Hz,

[0103] FIG. **15** is a flow chart illustrating a method **1500** of volume control reaching a maximum/minimum level on the audio output device **100**, according to some embodiments. Although the illustrated method **1500** includes specific steps, not all of the steps need to be performed or need to be performed in the order presented. In some embodiments, the method **1500** is executed by the audio output device controller **500**. The values of various thresholds or ranges described with respect to method **1500** are provided as non-limiting examples and other values for such parameters are possible and may vary based on the type or condition of the audio output device **100**, the type or condition of the external device **305**, other factors, or a combination thereof. [0104] The method **1500** includes the controller **500** determining that a volume actuator **175**, **180** is actuated for a predetermined amount of time (step **1505**). In some embodiments, the volume actuator **175**, **180** is actuated for a first predetermined amount of time. For example, the first predetermined amount of time may be any time less than 750 milliseconds (ms). In step **1510**, the controller **500** determines a requested volume level. In some embodiments, the controller **500** determines the requested volume level based on a number of times that the volume actuator **175** is actuated. For example, volume may be controlled in a 1:1 relationship with the volume selected on the external device **305** and a single actuation of the volume actuator **175**, **180** may increase the volume by one volume level. The minimum volume may be 0 and the maximum volume may be 15 such that 15 actuations of the volume increase actuator **175** would take the volume from 0 to 15. [0105] In step **1515**, the controller **500** provides a thirteenth light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the thirteenth light sequence. In some embodiments, the thirteenth light sequence is dependent on the requested volume level and is shown in Table 2, below.

TABLE-US-0002 TABLE 2 Thirteenth Light Sequence Volume Level Thirteenth Light Sequence

0 All segments 200, 205, 210, 215, 220 turned OFF 1 First segment 200 illuminated to a 30% illumination level 2 First segment 200 illuminated to a 60% illumination level 3 First segment 200 fully illuminated (100% illumination level) 4 First segment 200 fully illuminated, second segment 205 illuminated to a 30% illumination level 5 First segment 200 fully illuminated, second segment 205 illuminated to a 60% illumination level 6 First segment 200 fully illuminated, second segment 205 fully illuminated 7 First segment 200 fully illuminated, second segment 205 fully illuminated, third segment 210 illuminated to the 30% illumination level 8 First segment 200 fully illuminated, second segment 205 fully illuminated, third segment 210 illuminated to the 60% illumination level 9 First segment 200 fully illuminated, second segment 205 fully illuminated, third segment 210 fully illuminated 10 First segment 200 fully illuminated, second segment 205 fully illuminated, third segment 210 fully illuminated, fourth segment 215 illuminated to the 30% illumination level 11 First segment 200 fully illuminated, second segment 205 fully illuminated, third segment 210 fully illuminated, fourth segment 215 illuminated to the 60% illumination level 12 First segment 200 fully illuminated, second segment 205 fully illuminated, third segment 210 fully illuminated, fourth segment 215 fully illuminated 13 First segment 200 fully illuminated, second segment 205 fully illuminated, third segment 210 fully illuminated, fourth segment 215 fully illuminated, fifth segment 220 illuminated to the 30% illumination level 14 First segment 200 fully illuminated, second segment 205 fully illuminated, third segment 210 fully illuminated, fourth segment 215 fully illuminated, fifth segment 220 illuminated to the 60% illumination level 15 First segment 200 fully illuminated, second segment 205 fully illuminated, third segment 210 fully illuminated, fourth segment 215 fully illuminated, fifth segment 220 fully illuminated

[0106] In step **1520**, the controller **500** determines that the volume level is at a maximum/minimum level. For example, the controller may determine that the volume level is at 15 for a maximum level and at 0 for a minimum level. In step **1525**, the controller outputs an eleventh tone to the speaker **130**. In some embodiments, the eleventh tone is a certain pitch and frequency. For example, the eleventh tone may be two distinct sounds each a second length in time at the same pitch or frequency. In some embodiments, the eleventh tone is volume independent. In step **1530**, the controller **500** provides a fourteenth light sequence to be emitted by the light emitting unit **150**. In other words, the controller **500** operates the light emitting unit **150** according to the fourteenth light sequence. In some embodiments, the fourteenth light sequence includes flashing each segment **200, 205, 210, 215, 220** once at 0.5 Hz. When maximum volume is reached, the controller **500** then flashes each segment **200, 205, 210, 215, 220** at 4 Hz for 500 ms, twice.

[0107] FIG. **16** is a flow chart illustrating a method **1600** of controlling a volume and providing a tone in response to an actuation on the audio output device **100**, according to some embodiments. Although the illustrated method **1600** includes specific steps, not all of the steps need to be performed or need to be performed in the order presented. In some embodiments, the method **1600** is executed by the audio output device controller **500**. The values of various thresholds or ranges described with respect to method **1600** are provided as non-limiting examples and other values for such parameters are possible and may vary based on the type or condition of the audio output device **100**, the type or condition of the external device **305**, other factors, or a combination thereof.

[0108] The method **1600** includes the controller **500** determining that the audio output device **100** is outputting content over the speaker **130** (step **1605**). In step **1610**, the controller **500** determines that an actuator **170, 175, 180, 185, 190** is actuated. In step **1615**, the controller **500** decreases the volume of the content that is output over the speaker **130**. In some embodiments, the controller **500** drops the content from being played at volume level to a second volume level that is 50% of the volume level. In step **1620**, the controller **500** outputs a twelfth tone to the speaker **130**. The twelfth tone is dependent on the actuator that is actuated and may be any of the tones discussed herein. In some embodiments, the controller **500** plays the twelfth tone for a predetermined amount of time at a volume different from the volume level. In some embodiments, the predetermined amount of time

is 0.4 seconds. In some embodiments, all tones are played at volume level 8. In step 1625, the controller 500 increases the volume of the content that is output over the speaker 130. In some embodiments, the controller 500 gradually increases the volume level of the content back to the volume level once the tone is played for the predetermined amount of time. For example, the controller 500 may linearly or incrementally increase the volume level of the content. Alternatively, the controller 500 may exponentially or abruptly increase the volume level of the content.

[0109] FIG. 17 is a graph 1700 illustrating volume control when a tone is provided on the audio output device 100, according to some embodiments. As described above in the method 1600 of FIG. 16, when the volume increase actuator 175 is actuated, the controller 500 determines whether a maximum volume level is requested. When it is determined that the speaker 130 is outputting content at the maximum volume level and the volume increase actuator 175 is actuated, the controller 500 provides a tone to be played over the speaker 130. The controller 500 drops the content from being played at the maximum volume to a 50% volume output and plays the tone for a predetermined amount of time at a volume different from the maximum volume. In some embodiments, the predetermined amount of time is 0.4 seconds. In some embodiments, the maximum volume is a volume level 15 and the volume different from the maximum volume is a volume level 8. In some embodiments, all tones are played at volume level 8. The controller 500 may then gradually increase the volume level of the content back to the maximum volume level (e.g., volume level 15) once the tone is played for the predetermined amount of time. In some embodiments, the audio output device 100 plays content at a last volume level when the audio output device 100 was powered OFF.

[0110] In addition to playing audio files, the audio output device 100 may play phone calls over the speaker 130. In some embodiments, incoming phone calls may be muted and content playing over the speaker 130 may be paused. When a phone call ends, the content will ramp up to a selected volume (e.g., the volume at which the content was previously playing) and resume playing.

[0111] In some embodiments, the audio output device 100 may be reset to factory settings. For example, when the audio output device 100 is in a Bluetooth® pairing mode and the controller 500 determines that the Bluetooth® actuator 185, the daisy chain actuator 190, and the play actuator 170 are each actuated for 5 seconds, the controller 500 performs a factory settings reset. In some embodiments, the factory setting reset resets the memory 525 of the controller 500.

[0112] In some embodiments, when the audio output device 100 is powered ON and a charging cable is coupled to the charging port 195, the audio output device 100 will receive charging power. In some embodiments, when the audio output device 100 is powered OFF and a charging cable is coupled to the charging port 195, the audio output device 100 will not receive charging power.

[0113] In some embodiments, after a predetermined amount of time is elapsed, the controller 500 will control the light emitting unit 150 to dim the light segments 200, 205, 210, 215, 220 to a 50% illumination level. For example, the predetermined amount of time may be 3 minutes. The controller 500 may control the light segments 200, 205, 210, 215, 220 to fully illuminate (e.g., 100% illumination level) when an input from any actuator is received.

[0114] The embodiments described above and illustrated in the figures are presented by way of example only and are not intended as a limitation upon the concepts and principles of the present invention. As such, it will be appreciated that various changes in the elements and their configuration and arrangement are possible without departing from the spirit and scope of the present invention. For example, although specific light sequences are described for particular functions of the audio output device 100, the light emitting unit 150 may emit other sequences or patterns of light for each function or for different functions. The sequences may include various combinations of turning on, turning off, flashing, dimming, intensifying, etc. the light segments 200, 205, 210, 215, 220.

[0115] Various features and advantages of the invention are set forth in the following claims.

Claims

1. An audio output device comprising: a housing; a speaker provided in the housing; a plurality of actuators provided on the housing; a light emitting unit provided on the housing, the light emitting unit including a plurality of light emitting segments arranged in a line; and a controller located within the housing and including an electronic processor, a memory, and a transceiver, the controller coupled to the speaker, the plurality of actuators, and the light emitting unit, the controller configured to: determine that a first actuator of the plurality of actuators is actuated, and operate the light emitting unit according to a first light sequence in response to the first actuator being actuated.
2. The audio output device of claim 1, wherein a first light emitting segment of the plurality of light emitting segments includes a first LED and a second light emitting segment of the plurality of light emitting segments includes a second LED, wherein the first LED is illuminated at a first brightness and the second LED is illuminated at a second brightness during the first light sequence, and wherein the first brightness is different than the second brightness.
3. The audio output device of claim 1, wherein a first light emitting segment of the plurality of light emitting segments includes a first LED and a second light emitting segment of the plurality of light emitting segments includes a second LED, wherein the first LED is illuminated at a first time and the second LED is illuminated at a second time during the first light sequence, and wherein the second time is after the first time.
4. The audio output device of claim 1, wherein the first light sequence indicates initiation of an operation of the audio output device, and wherein the controller is also configured to operate the light emitting unit according to a second light sequence upon completion of the operation.
5. The audio output device of claim 4, wherein the second light sequence is different than the first light sequence.
6. The audio output device of claim 1, wherein the controller is configured to: determine that a second actuator of the plurality of actuators is actuated, and operate the light emitting unit according to a second light sequence in response to the second actuator being actuated, the second light sequence being different than the first light sequence.
7. The audio output device of claim 6, wherein the first actuator initiates pairing of the audio output device with an external device that provides content to the audio output device, and wherein the second actuator initiates pairing of the audio output device with another device that receives content from the audio output device.
8. The audio output device of claim 1, wherein the controller is also configured to output a tone over the speaker in response to the first actuator being actuated.
9. The audio output device of claim 8, wherein the tone is synched with the first light sequence.
10. The audio output device of claim 8, wherein the controller is also configured to drop a volume level of content being output by the speaker while outputting the tone.
11. The audio output device of claim 10, wherein the controller is also configured to gradually increase the volume level of the content being output by the speaker after the tone is played.
12. The audio output device of claim 1, wherein each actuator of the plurality of actuators includes an LED that illuminates upon actuation.
13. The audio output device of claim 1, wherein the housing is an elongated housing having a first end, a second opposite the first end, and a longitudinal axis extending through the first end and second end, and wherein the plurality of light emitting segments is arranged in the line extending parallel to the longitudinal axis.
14. The audio output device of claim 13, wherein the plurality of actuators is also arranged in a line extending parallel to the longitudinal axis.
15. The audio output device of claim 1, further comprising a battery pack interface configured to

receive a battery pack.

16. An audio output device comprising: a housing; a speaker provided in the housing; a plurality of actuators provided on the housing; a light emitting unit provided on the housing, the light emitting unit including a plurality of light emitting segments arranged in a line; and a controller located within the housing and including an electronic processor, a memory, and a transceiver, the controller coupled to the speaker, the plurality of actuators, and the light emitting unit, the controller configured to: determine that a first actuator of the plurality of actuators is actuated to initiate an operation, operate the light emitting unit according to a first light sequence in response to the first actuator being actuated and the operation being initiated, and operate the light emitting unit according to a second light sequence in response to the operation being completed, the second light sequence being different than the first light sequence.

17. The audio output device of claim 16, wherein during the first light sequence, the plurality of light emitting segments is illuminated in a first order, and wherein during the second light sequence, the plurality of light emitting segments is illuminated in a second order that is different than the first order.

18. The audio output device of claim 16, wherein the controller is also configured to: determine that a second actuator of the plurality of actuators is actuated, and operate the light emitting unit according to a third light sequence in response to the second actuator being actuated, the third light sequence being different than the first light sequence and the second light sequence.

19. An audio output device comprising: an elongated housing having a first end, a second end opposite the first end, and a longitudinal axis extending through the first end and the second end; a speaker provided in the elongated housing; a plurality of actuators provided on the elongated housing; and a light emitting unit provided on the elongated housing and including a plurality of light emitting segments arranged in a line extending parallel to the longitudinal axis, the light emitting unit operable to emit a light sequence in response to actuation of one of the plurality of actuators.

20. The audio output device of claim 19, wherein each light emitting segment of the plurality of light emitting segments corresponds to one actuator of the plurality of actuators.

21. The audio output device of claim 19, wherein the plurality of light emitting segments is arranged sequentially on the elongated housing without interruption.

22. The audio output device of claim 19, wherein the plurality of actuators is also arranged in a line extending parallel to the longitudinal axis.

23. The audio output device of claim 22, wherein the plurality of light emitting segments is next to the plurality of actuators.

24. The audio output device of claim 19, wherein the first end of the elongated housing includes a battery pack interface configured to receive a battery pack, and wherein the second end of the elongated housing is configured to support the audio output device.

25. The audio output device of claim 19, further comprising a fuel gauge provided on the elongated housing.
